

DTSA 5505

Data Mining Methods

FORMAT	CATEGORY	CREDITS
Online (DTSA)	Core	1
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 3 – Core	Data Scientist (46.3 / 100)	32.6 / 100
SPECIALIZATION Data Mining Foundations and Practice		

33

CURRENT AVG SCORE
across 6 job roles

77

POTENTIAL SCORE
with recommended additions

+69

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

This course covers core techniques used in data mining, including frequent pattern analysis, classification, clustering, outlier detection, as well as time-series mining and graph mining.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
Data Scientist	46.3	87.2	+40.9	HIGH
ML Engineer	41.7	78.9	+37.2	HIGH
Machine Learning Engineer	38.9	82.5	+43.5	HIGH
AI Engineer	29.1	65.8	+36.7	HIGH
Applied Scientist	22.6	92.0	+69.4	HIGH
MLOps Engineer	16.8	58.1	+41.3	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (example) appear as gaps across 3 or more job roles.

System Design 4 of 6 roles

systems

Feature Engineering 1 of 6 roles

preprocessing

Business Communication 1 of 6 roles

business

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1 **HIGH** **Add System Design coverage** — required vocabulary in 4 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.

2 **LOW** **Optional: Feature Engineering** — would broaden appeal to one additional role. Candidate terms: preprocessing.

3 **LOW** **Optional: Business Communication** — would broaden appeal to one additional role. Candidate terms: business.

GAP KEYWORDS BY ROLE

Data Scientist

46.3 → 87.2 (+40.9 pts)

toolkit sets analytics social

ML Engineer

41.7 → 78.9 (+37.2 pts)

discovery learning machine users systems models impact digital

Machine Learning Engineer

38.9 → 82.5 (+43.5 pts)

web sets learning analytics tools business mobile design

AI Engineer

29.1 → 65.8 (+36.7 pts)

following models learning software systems technology machine information

Applied Scientist

22.6 → 92.0 (+69.4 pts)

discovery scientific research literature models learning systems industry

MLOps Engineer

16.8 → 58.1 (+41.3 pts)

text systems learning machine preprocessing image tools models

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.